



In sync: Girish Venkatasubban (left) and Ganesh Krish, the duo behind the development of an indigenous Indian compound bow. SPECIAL ARRANGEMENT

compound events. Thousands of archers compete with compound bows domestically. Every one of them used bows imported from China or the USA. Girish wondered if he could change that.

"At our unit, we produce parts that go into rockets and missile systems where the tolerances are measured in microns, and there's next to no margin for failure. If we can do that, why can't we make a bow?"

A few weeks later, Imaxx Sports was founded out of the same Microfine Bushings factory in Chennai, with a few engineers from the parent company joining the effort.

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Initially, they decided to reverse-engineer what already existed.

"We took an imported compound bow. We separated its components. We measured everything. We got a

mould made from that and cast our first bow in aluminium," he says.

He took the bow to a local competition in Chennai and, on the advice of a coach, showed the product to Ganesh Krish.

Ganesh is not a manufacturer. He is a commerce graduate who developed a keen interest in compound archery while taking a break from work in Texas, USA. He immersed himself in understanding the engineering side of compound bow-making and archery. Within India's domestic archery circuit, and even among members of the national squad, Ganesh had developed a reputation for understanding archery and bow setups at a highly technical and scientific level.

"There was a coach who told me I needed to see this bow that Girish had brought, and when I saw it, I was intrigued," says Krish.

On the surface, a compound bow looks straightforward enough to replicate. It comprises a curved frame, a pulley system to reduce the difficulty of holding the bow, and a string. That simplicity, though, is misleading.

"A professional compound bow must deliver identical performance across repeated shots. There are some 44 components in a compound bow, including the riser, cams, limbs and strings. Each has a unique purpose. The riser forms the foundation for shot stability. The cams are designed to distribute energy at a specific point and also reduce the effort required to hold the bow at full draw. The limbs must flex and recover within controlled limits to generate energy, and that energy must then be transferred to the arrow through the string with high efficiency. Everything must work together," he says.

Although the first bow was imperfect, Krish says he saw enough potential in the venture to eventually join Girish on the project.

For both men, this was unfamiliar territory.

"It wasn't just a matter of precision. When you work in aerospace, you know exactly what the standards are. But because no one had ever made a compound bow in India, we had to figure out the standards ourselves. We had to define what mattered," Girish says.

The two eventually began breaking down exactly what their bow needed to achieve.

"We needed to know how fast the arrow travels. How tightly do the arrows need to group? How does the load build while drawing the string? Even something as simple as measurements was not straightforward because, in India, we use the metric system, while compound bows originate from the USA, where everything is measured in pounds and inches. We had to start creating our own framework," Krish says.

None of this was easy. Or cheap.

Prototype after prototype had to be machined out of aluminium. Designs were tested, rejected and redesigned. Small changes required new parts. Entire assemblies were scrapped.

"We went through 30 to 40 aluminium blocks. Each one costs money. I think I would have spent around Rs. 50 lakh on prototypes and